Ethan Bikhazi-Green

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EDUCATION

University of California, Santa Barbara EXPECTED JUN 2027
Bachelor of Science (B.S.), Mechanical Engineering
Relevant Coursework: Finite Element Analysis, Controls, Mechatronics, Dynamical Systems, Machine Learning, Robotics.

SKILLS

CAD/Manufacturing: SolidWorks, Fusion 360, FEA, CAM, Machining, CNC Milling, Lathe, 3D Printing, GD&T.
Programming/Technical: MATLAB, Simulink, Python, Arduino, KiCad, Power Automate, LabVIEW, Microsoft Office.

EXPERIENCE

- FARM CREDIT EAST. New York - *Business Automation & AI Solutions Intern* JUN 2026-PRESENT
  - Developed and deployed AI agents and automated workflows using Microsoft Power Platform, Copilot Studio, Azure, and SharePoint, generating **3,300+ employee-hours** of annual operational capacity.
  - Developed an HR AI agent designed for **500+ employees**, integrating **40,000+ documents** while testing failure cases and guardrails for consistent, safe responses.
- ATOM-IQ. Santa Barbara - *Undergraduate Research Assistant* JAN 2026-MAY 2026
  - Designed and built **$40K+** in aluminum-extrusion gantry infrastructure to support instrumentation for an autonomous materials-testing laboratory.
  - Coordinated mechanical mounting and integration requirements across **60+** undergraduate contributors and 5 teams.
- GAUCHO ROCKET PROJECT. Santa Barbara - *Aerospace Engineer* AUG 2024-JUN 2025
  - Modeled PID/TVC control systems using Simulink and Python to target **±3° stability** and **<1 m/s landing velocity.**
  - Designed and fabricated rocket fins using CAD and 3D printing to improve aerodynamic stability and flight performance
  - Created and interpreted SolidWorks engineering drawings using ASME Y14.5 GD&T to specify component tolerances and guide fabrication and assembly.

PROJECTS

- STROKE FALL PREVENTION WEARABLE. *Senior Capstone, Lead Engineer* SEP 2026-PRESENT
  - Leading engineering design for a 5-person technical team developing a wearable obstacle-warning system for stroke patients, overseeing prototyping and validation planning. Managing a **$6,000** budget.
  - Developed CAD models and physical prototypes to evaluate sensor placement, ergonomics, and enclosure design; designed a custom PCB for a primary prototype.
  - Evaluating 14 prototype concepts across LiDAR, ultrasonic sensing, AI-based computer vision, haptic and audio feedback, and attachment mechanisms; developing quantitative test protocols informed by 10+ clinical and user interviews.
- FRISBEE LAUNCHER. *Design Competition* MAR 2026
  - Designed and fabricated a motorized frisbee launcher achieving **20 m/s** launch velocity, 50-yard range, and **>99%** successful launches within a $200 budget.
  - Sized the motor and flywheel using first-principles calculations and MATLAB modeling; refined the mechanism through iterative prototyping and slow-motion video analysis.
  - Designed launcher circuitry in KiCad, sourced and assembled electronic components, and fabricated a 3D-printed, vibration-resistant enclosure.
- LUG WRENCH. *Structural Analysis & Redesign* JAN 2026
  - Redesigned an AISI 1045 lug wrench using SolidWorks FEA, reducing peak von Mises stress by **39%** to meet yield-strength requirements.
  - Applied localized mesh refinement and convergence analysis to improve FEA reliability at critical stress concentrations.
- AIR MOTOR. *Machine Shop* MAY 2025
  - Machined and assembled a **5,000 RPM** air motor using CNC milling and lathe operations.
- INDEPENDENT POKER PLAYER. DEC 2023-PRESENT
  - Generated **$77K+** in live/online profit at an $84/hour win rate; managed **$20K+** in investor-backed capital.